

ALY 6010 Prob Theory and Intro Stat – Report

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Module 6 R Practice

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Introduction :

The dataset analyzed in this report comprises 9358 instances of hourly averaged responses recorded by a gas multi sensor device deployed in an urban environment within an Italian city. Collected over the course of one year, from March 2004 to February 2005, The primary objective of this report is to analyze the relationship between various predictor variables and the concentration of nitrogen dioxide (NO₂) in the air. Specifically, two models will be fitted to the NO₂ concentration data, each employing a different approach to specify predictor variables. Overall, this analysis aims to provide insights into the factors influencing NO₂ concentration levels in urban environments, leveraging the rich dataset of hourly averaged responses from the multi sensor device.

Problem statement:

The problem is figuring out the exact levels of pollution in cities using data from special sensors. Even though we have accurate pollution readings from a standard device, issues like sensor errors and missing data make it hard to trust the results from these special sensors. This makes it tough to understand and fix air pollution problems in cities. We need better ways to analyze and understand this data so we can tackle pollution more effectively.

Installing libraries

```
library(tidyverse)

## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    3.4.4      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.0
## ✓ purrr      1.0.2
## — Conflicts ————— tidyverse_conflict
s() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors

library(ggplot2)
library(caret)

## Loading required package: lattice
##
## Attaching package: 'caret'
##
```

```
## The following object is masked from 'package:purrr':
##
##   lift

library(Metrics)

##
## Attaching package: 'Metrics'
##
## The following objects are masked from 'package:caret':
##
##   precision, recall

library(VIM)

## Loading required package: colorspace
## Loading required package: grid
## VIM is ready to use.
##
## Suggestions and bug-reports can be submitted at: https://github.com/statistikat/VIM/issues
##
## Attaching package: 'VIM'
##
## The following object is masked from 'package:datasets':
##
##   sleep

library(corrplot)

## corrplot 0.92 loaded

library(GGally)

## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg      ggplot2

library(MASS)

##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##   select
```

Data Used:

Reading dataset : Importing the file on dataset in R and reading dataset

```
Air_quality <- read.csv("AirQualityUCI.csv")
head(Air_quality)
```

##	Date	Time	CO.GT.	PT08.S1.CO.	NMHC.GT.	C6H6.GT.	PT08.S2.NMHC.	NOx.GT.
## 1	10/3/2004	18.00.00	2.6	1360	150	11.9	1046	166
## 2	10/3/2004	19.00.00	2.0	1292	112	9.4	955	103
## 3	10/3/2004	20.00.00	2.2	1402	88	9.0	939	131
## 4	10/3/2004	21.00.00	2.2	1376	80	9.2	948	172
## 5	10/3/2004	22.00.00	1.6	1272	51	6.5	836	131
## 6	10/3/2004	23.00.00	1.2	1197	38	4.7	750	89

##	PT08.S3.NOx.	NO2.GT.	PT08.S4.NO2.	PT08.S5.O3.	T	RH	AH
## 1	1056	113	1692	1268	13.6	48.9	0.7578
## 2	1174	92	1559	972	13.3	47.7	0.7255
## 3	1140	114	1555	1074	11.9	54.0	0.7502
## 4	1092	122	1584	1203	11.0	60.0	0.7867
## 5	1205	116	1490	1110	11.2	59.6	0.7888
## 6	1337	96	1393	949	11.2	59.2	0.7848

Columns presented in dataset "Air Quality"

```
colnames(Air_quality) <- c("Date", "Time", "CO.GT", "PT08.S1.CO", "NMHC.GT", "C6H6.GT", "PT08.S2.NMHC", "NOx.GT", "PT08.S3.NOx", "NO2.GT", "PT08.S4.NO2", "PT08.S5.O3", "T", "RH", "AH")
colnames(Air_quality)
```

## [1]	"Date"	"Time"	"CO.GT"	"PT08.S1.CO"	"NMHC.GT"
## [6]	"C6H6.GT"	"PT08.S2.NMHC"	"NOx.GT"	"PT08.S3.NOx"	"NO2.GT"
## [11]	"PT08.S4.NO2"	"PT08.S5.O3"	"T"	"RH"	"AH"

Descriptive Statistics

Descriptive statistics provide a summary of the characteristics of a dataset, helping to understand its distribution and key features. In the function `summary()`, provides descriptive statistics for the variables in the "Air Quality" dataset.

```
summary(Air_quality)
```

##	Date	Time	CO.GT	PT08.S1.CO
##	Length:9357	Length:9357	Min. :-200.00	Min. :-200
##	Class :character	Class :character	1st Qu.: 0.60	1st Qu.: 921

```
## Mode :character Mode :character Median : 1.50 Median :1053
## Mean : -34.03 Mean :1049
## 3rd Qu.: 2.60 3rd Qu.:1221
## Max. :1428.00 Max. :2040
##
## NMHC.GT C6H6.GT PT08.S2.NMHC NOx.GT
## Min. : -200.0 Min. : -200.000 Min. : -200.0 Min. : -200.0
## 1st Qu.: -200.0 1st Qu.: 4.000 1st Qu.: 711.0 1st Qu.: 50.0
## Median : -200.0 Median : 7.900 Median : 895.0 Median : 141.0
## Mean : -159.1 Mean : 2.005 Mean : 894.4 Mean : 168.7
## 3rd Qu.: -200.0 3rd Qu.: 13.600 3rd Qu.:1105.0 3rd Qu.: 284.0
## Max. :1189.0 Max. :1321.000 Max. :2214.0 Max. :1479.0
##
## PT08.S3.NOx NO2.GT PT08.S4.NO2 PT08.S5.O3
## Min. : -200.0 Min. : -200.00 Min. : -200 Min. : -200.0
## 1st Qu.: 637.0 1st Qu.: 53.00 1st Qu.:1185 1st Qu.: 700.0
## Median : 794.0 Median : 96.00 Median :1446 Median : 942.0
## Mean : 794.9 Mean : 58.36 Mean :1391 Mean : 974.9
## 3rd Qu.: 960.0 3rd Qu.: 133.00 3rd Qu.:1662 3rd Qu.:1255.0
## Max. :2683.0 Max. :1779.00 Max. :2775 Max. :2523.0
##
## T RH AH
## Min. : -200.000 Min. : -200.00 Min. : -200.0000
## 1st Qu.: 10.900 1st Qu.: 34.10 1st Qu.: 0.6923
## Median : 17.200 Median : 48.50 Median : 0.9768
## Mean : 9.782 Mean : 39.48 Mean : -6.8385
## 3rd Qu.: 24.100 3rd Qu.: 61.90 3rd Qu.: 1.2963
## Max. : 57.200 Max. : 88.70 Max. : 2.2310
## NA's :1
```

Dataset structure

Structuring the dataset to examine its characteristics, including the variables it contains, their types, the number of observations, and any missing values or outliers.

```
str(Air_quality)

## 'data.frame': 9357 obs. of 15 variables:
## $ Date : chr "10/3/2004" "10/3/2004" "10/3/2004" "10/3/2004" ...
## $ Time : chr "18.00.00" "19.00.00" "20.00.00" "21.00.00" ...
## $ CO.GT : num 2.6 2 2.2 2.2 1.6 1.2 1.2 1 0.9 0.6 ...
## $ PT08.S1.CO : int 1360 1292 1402 1376 1272 1197 1185 1136 1094 1010 ..
## $ NMHC.GT : num 150 112 88 80 51 38 31 31 24 19 ...
## $ C6H6.GT : num 11.9 9.4 9 9.2 6.5 4.7 3.6 3.3 2.3 1.7 ...
## $ PT08.S2.NMHC: int 1046 955 939 948 836 750 690 672 609 561 ...
## $ NOx.GT : int 166 103 131 172 131 89 62 62 45 -200 ...
## $ PT08.S3.NOx : int 1056 1174 1140 1092 1205 1337 1462 1453 1579 1705 ..
## $ NO2.GT : int 113 92 114 122 116 96 77 76 60 -200 ...
## $ PT08.S4.NO2 : int 1692 1559 1555 1584 1490 1393 1333 1333 1276 1235 ..
```

```
.
## $ PT08.S5.O3 : num 1268 972 1074 1203 1110 ...
## $ T : num 13.6 13.3 11.9 11 11.2 11.2 11.3 10.7 10.7 10.3 ...
## $ RH : num 48.9 47.7 54 60 59.6 59.2 56.8 60 59.7 60.2 ...
## $ AH : num 0.758 0.726 0.75 0.787 0.789 ...
```

9357 Observations and 15 Variables

Data types:

Date, Time: Character

CO.GT, NMHC.GT, C6H6.GT, PT08.S5.O3, T, RH, AH : Numeric

PT08.S1.CO, PT08.S2.NMHC, NOx.GT, PT08.S3.NOx, NO2.GT, PT08.S4.NO2: Integer

Data Transformation and Cleaning

Filtering only numeric columns from the dataset "Air Quality"

Filtering only numeric columns to perform numerical analyses. By removing non-numeric columns, focusing on the variables relevant to analysis and ensuring compatibility with numbers based operations such statistical calculations and graphical representations.

```
Air_quality <- Air_quality[sapply(Air_quality, is.numeric)]
```

Check for missing values

Checking for missing values in a dataset is essential to facilitate data preprocessing and ensure data quality.

```
cat("Checking for missing values (NAs) in the dataset:\n")
## Checking for missing values (NAs) in the dataset:
sum(is.na(Air_quality))
## [1] 1
cat("\nNumber of missing values per column:\n")
##
## Number of missing values per column:
colSums(is.na(Air_quality))
##          CO.GT  PT08.S1.CO          NMHC.GT          C6H6.GT PT08.S2.NMHC          NOx
##          0          0          0          0          0          0
## PT08.S3.NOx          NO2.GT  PT08.S4.NO2  PT08.S5.O3          T
##          0          0          0          0          0          0
```

```
0
##      AH
##      1
```

The total number of missing values in the entire dataset "**AirQuality**", is found to be **1**. Present in "**AH**" column, while all other columns have no missing values.

Handle missing values (coded as -200 in this dataset)

Replacing all the values equal to -200 in the "AirQuality" dataset with NA (missing value). Making it easier to handle and analyze missing data in subsequent operations.

```
Air_quality[Air_quality == -200] <- NA
```

Impute missing values or drop them

For Simplifying this analysis, we remove rows with NAs

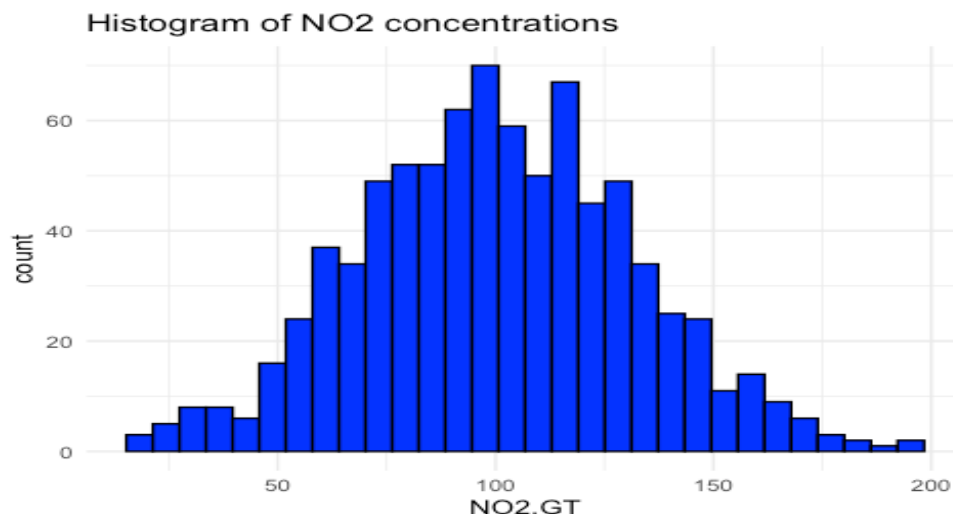
```
Air_quality <- na.omit(Air_quality)
```

Exploratory Data Analysis (EDA)

Conducting a thorough EDA to understand the distributions, relationships, and trends within the data. Using visualizations such as scatter plots and histograms to examine the relationships between the dependent and independent variables.

Histogram of NO2

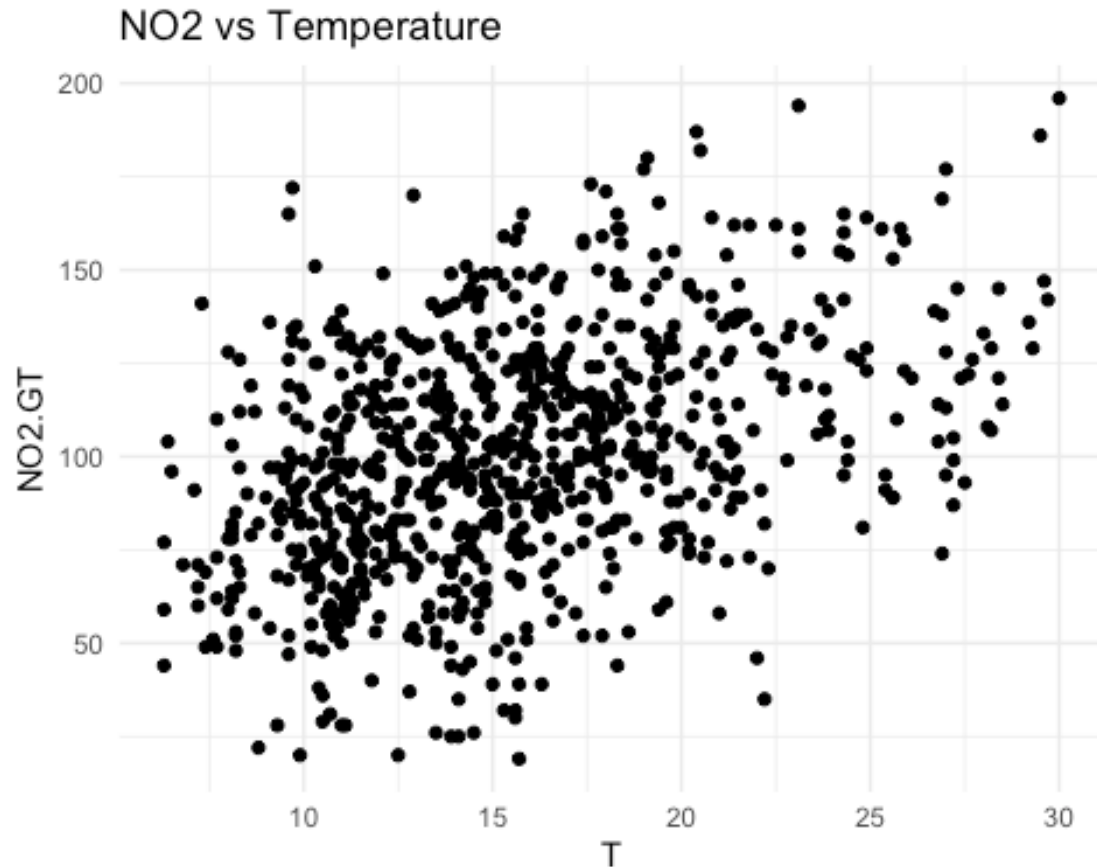
```
ggplot(Air_quality, aes(x=NO2.GT)) + geom_histogram(bins=30, fill="blue", col="black") +
  theme_minimal() + ggtitle("Histogram of NO2 concentrations")
```



The x-axis represents the NO2 concentrations, while the y-axis indicates the frequency of occurrence. The histogram reveals that the distribution of NO2 concentrations is positively skewed.

Scatter plot of NO2 vs Temperature

```
ggplot(Air_quality, aes(x = T, y = NO2.GT)) +
  geom_point() +
  ggtitle("NO2 vs Temperature") +
  theme_minimal()
```



The scatter plot of NO2 versus temperature illustrates the relationship between these two variables in the dataset. Each data point represents a specific observation, with the x-axis denoting the temperature and the y-axis representing the NO2.

Calculate the correlation matrix

```
correlation_matrix <- cor(Air_quality[, sapply(Air_quality, is.numeric)], use
  ="complete.obs")
correlation_matrix
```

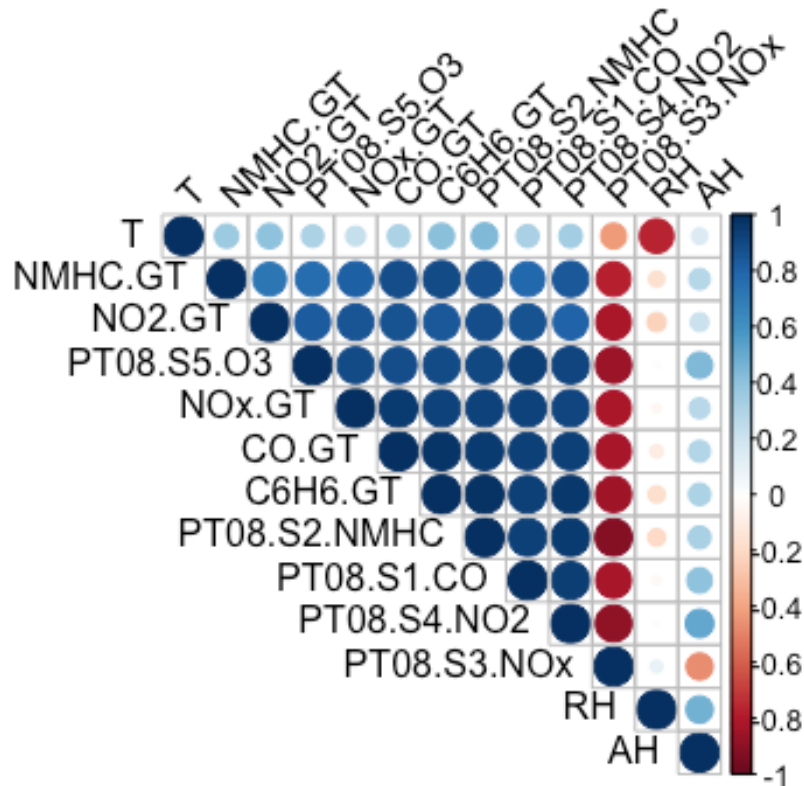
##	CO.GT	PT08.S1.CO	NMHC.GT	C6H6.GT	PT08.S2.NMHC
## CO.GT	1.0000000	0.93626072	0.8871675	0.9726596	0.9584256
## PT08.S1.CO	0.9362607	1.00000000	0.7817468	0.9313679	0.9363456
## NMHC.GT	0.8871675	0.78174678	1.00000000	0.8979281	0.8750609
## C6H6.GT	0.9726596	0.93136785	0.8979281	1.00000000	0.9848344

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##	PT08.S2.NMHC	0.9584256	0.93634564	0.8750609	0.9848344	1.0000000
##	NOx.GT	0.9513417	0.92288458	0.8111820	0.9273038	0.9266333
##	PT08.S3.NOx	-0.8237281	-0.82957680	-0.7742368	-0.8488501	-0.9106509
##	NO2.GT	0.8614319	0.86657945	0.7280519	0.8467426	0.8850232
##	PT08.S4.NO2	0.9399209	0.94502046	0.8484892	0.9608112	0.9578834
##	PT08.S5.O3	0.8829433	0.93501070	0.7619046	0.8969779	0.9091000
##	T	0.3182609	0.32481533	0.3669761	0.4184086	0.4456146
##	RH	-0.1051570	-0.03957004	-0.1602573	-0.1784103	-0.1933332
##	AH	0.2955910	0.40703803	0.2821422	0.3134152	0.3253333
##		NOx.GT	PT08.S3.NOx	NO2.GT	PT08.S4.NO2	PT08.S5.O3
##	CO.GT	0.95134166	-0.82372808	0.8614319	0.93992093	0.88294328
##	PT08.S1.CO	0.92288458	-0.82957680	0.8665794	0.94502046	0.93501070
##	NMHC.GT	0.81118197	-0.77423675	0.7280519	0.84848921	0.76190464
##	C6H6.GT	0.92730382	-0.84885013	0.8467426	0.96081124	0.89697791
##	PT08.S2.NMHC	0.92663333	-0.91065086	0.8850232	0.95788342	0.90910003
##	NOx.GT	1.00000000	-0.81429650	0.8574250	0.91272368	0.89338083
##	PT08.S3.NOx	-0.81429650	1.00000000	-0.8152242	-0.88021333	-0.85752570
##	NO2.GT	0.85742501	-0.81522418	1.00000000	0.80779158	0.83965616
##	PT08.S4.NO2	0.91272368	-0.88021333	0.8077916	1.00000000	0.91899557
##	PT08.S5.O3	0.89338083	-0.85752570	0.8396562	0.91899557	1.00000000
##	T	0.23839487	-0.42394604	0.4068066	0.34362641	0.31345790
##	RH	-0.04197541	0.09683373	-0.2230332	0.01238939	-0.01452952
##	AH	0.27067928	-0.46301585	0.2145590	0.51553031	0.44019790
##		T	RH	AH		
##	CO.GT	0.3182609	-0.10515700	0.2955910		
##	PT08.S1.CO	0.3248153	-0.03957004	0.4070380		
##	NMHC.GT	0.3669761	-0.16025735	0.2821422		
##	C6H6.GT	0.4184086	-0.17841035	0.3134152		
##	PT08.S2.NMHC	0.4456146	-0.19333324	0.3253333		
##	NOx.GT	0.2383949	-0.04197541	0.2706793		
##	PT08.S3.NOx	-0.4239460	0.09683373	-0.4630158		
##	NO2.GT	0.4068066	-0.22303318	0.2145590		
##	PT08.S4.NO2	0.3436264	0.01238939	0.5155303		
##	PT08.S5.O3	0.3134579	-0.01452952	0.4401979		
##	T	1.0000000	-0.76986905	0.1599644		
##	RH	-0.7698691	1.00000000	0.4757759		
##	AH	0.1599644	0.47577590	1.0000000		

Visualize the correlation matrix

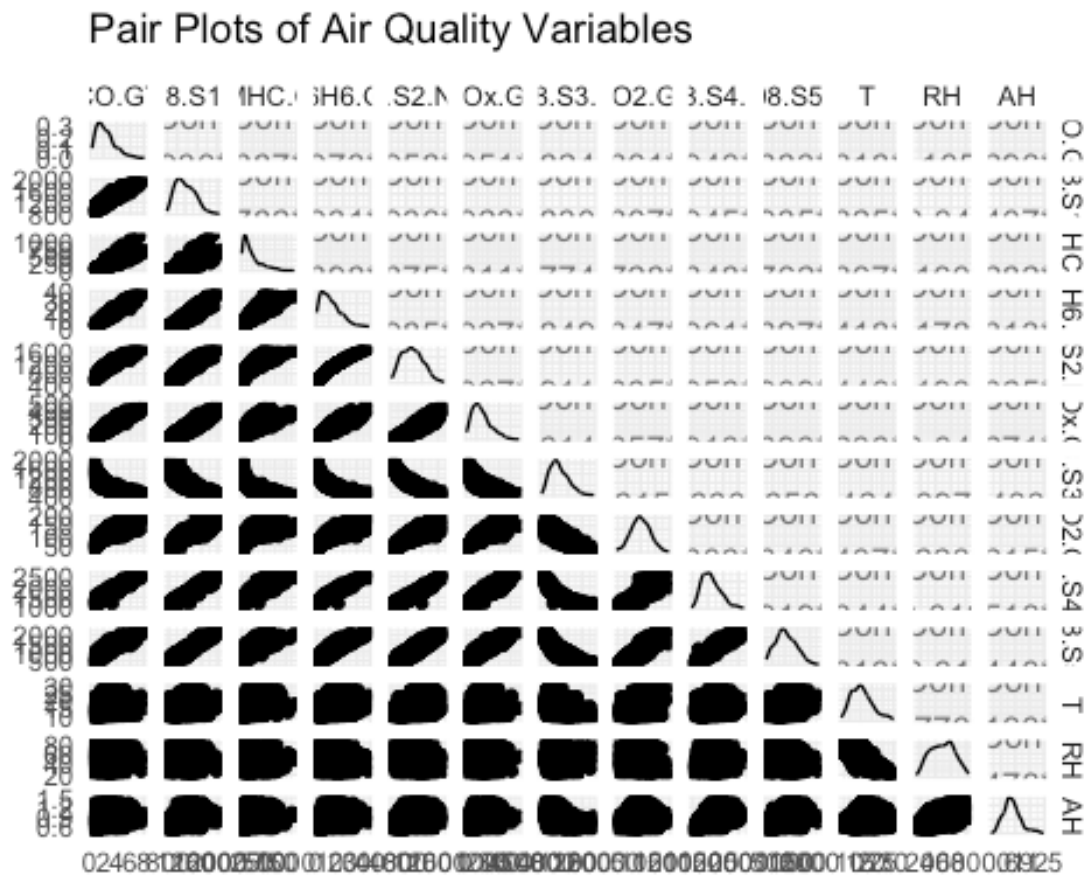
```
corrplot(correlation_matrix, method="circle", type="upper", order="hclust",
         tl.col="black", tl.srt=45,
         title="Correlation Matrix of Air Quality Variables")
```

Correlation Matrix of Air Quality Variables

CO.GT, PT08.S1.CO, C6H6.GT, PT08.S2.NMHC, and NOx.GT shows strong positive correlations with each other, indicates that if one variables tend to increase the other increases too. PT08.S3.NOx and RH have a moderately strong negative correlation, which means they typically move in opposite directions. But, PT08.S4.NO2 and AH also shows a moderately strong positive correlation.

Generate pair plots

```
ggpairs(Air_quality[, sapply(Air_quality, is.numeric)]) +
  ggtitle("Pair Plots of Air Quality Variables") +
  theme_minimal()
```



This plot shows numeric presentation of variables in dataset. Each small plots shows pair relationship in between variables.

Modeling setting seed For reproducibility

Ensuring reproducibility in random processes for training and testing in data set.

```
set.seed(123) # For reproducibility
```

Split the data into training and testing sets

Splitting the data into training and testing sets to train our model on one subset (the training set) and evaluate its performance on another independent subset (the testing set).

```
trainIndex <- createDataPartition(Air_quality$NO2.GT, p=.8, list=FALSE, times
=1)
trainData <- Air_quality[trainIndex, ]
testData <- Air_quality[-trainIndex, ]
```

Fit the linear regression model (MODEL-1)

```
modelLm <- lm(NO2.GT ~ ., data=trainData)
summary(modelLm)

##
## Call:
## lm(formula = NO2.GT ~ ., data = trainData)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -46.728  -6.957   0.495   7.918  42.533
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -3.826735   23.376474  -0.164   0.87002
## CO.GT         11.538104    1.826824   6.316 4.98e-10 ***
## PT08.S1.CO     0.046268    0.008517   5.433 7.86e-08 ***
## NMHC.GT       -0.003029    0.005594  -0.541  0.58836
## C6H6.GT       -3.357404    0.741664  -4.527 7.12e-06 ***
## PT08.S2.NMHC   0.200256    0.021684   9.235 < 2e-16 ***
## NOx.GT         0.057282    0.020823   2.751  0.00611 **
## PT08.S3.NOx    0.006093    0.007369   0.827  0.40860
## PT08.S4.NO2   -0.126291    0.013748  -9.186 < 2e-16 ***
## PT08.S5.O3     0.015034    0.003728   4.033 6.16e-05 ***
## T              1.059955    0.440180   2.408  0.01632 *
## RH             0.297593    0.152595   1.950  0.05158 .
## AH             6.993635   10.570615   0.662  0.50846
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.29 on 650 degrees of freedom
## Multiple R-squared:  0.8747, Adjusted R-squared:  0.8724
## F-statistic: 378.1 on 12 and 650 DF, p-value: < 2.2e-16
```

The summary() function provides coefficients and their significance

The summary() function provides coefficients and their significance levels for each predictor variable in the linear regression model.

Interpretation of Model Parameters and Fitness

The linear regression model fitted to the data predicts the concentration of Nitrogen Dioxide (NO2.GT) based on several predictor variables representing different air quality parameters measured by chemical sensors.

Coefficients: The coefficients indicate the estimated change in NO₂ concentration for a one-unit increase in each predictor variable, holding other variables constant. If CO.GT compared to NO₂ concentration, it can be said that if one-unit increase in CO.GT (Carbon Monoxide concentration) is then 11.54 units increases in NO₂ concentration.

Intercept: The intercept (-3.83) represents the estimated NO₂ concentration when all predictor variables are zero.

P-value: The p-values associated with each coefficient indicate the significance of each predictor variable in predicting NO₂ concentration.

Adjusted R-squared: 0.8724 indicates that approximately 87.24% of the variance in NO₂ concentration can be explained by the predictor variables included in the model.

However, it's important to note that some predictor variables, such as NMHC.GT (Non-Methane Hydrocarbons), PT08.S3.NO_x, and AH (Absolute Humidity), have p-values greater than 0.05, indicating that they may not be statistically significant predictors of NO₂ concentration in this model.

Evaluation Metrics

```
predictions <- predict(modelLm, testData)
rmse_val <- rmse(testData$NO2.GT, predictions)
rsquared_val <- R2(testData$NO2.GT, predictions)

cat("RMSE: ", rmse_val, "\n")

## RMSE: 12.33545

cat("R-squared: ", rsquared_val, "\n")

## R-squared: 0.8455459
```

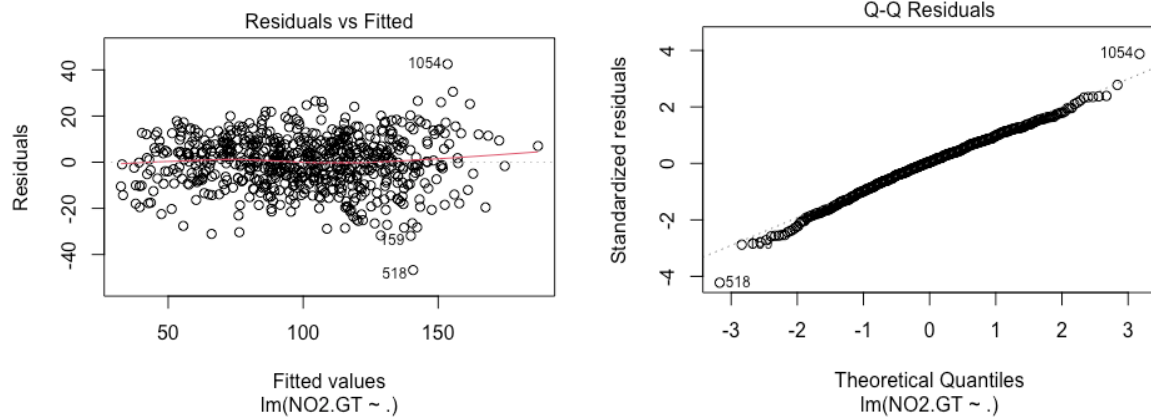
RMSE (Root Mean Squared Error) and R-squared, to assess the performance of the linear regression model on the test dataset.

RMSE (Root Mean Squared Error): It measures the average magnitude of the errors between predicted and observed values. A lower RMSE indicates better model performance, with 12.33545 being the computed value in this case.

R-squared (Coefficient of Determination): It represents the proportion of variance in the dependent variable (NO₂ concentration) that is explained by the independent variables in the model. R-squared values range from 0 to 1, with higher values indicating better fit. In this case, an R-squared value of approximately 0.846 suggests that the model accounts for about 84.65% of the variability in NO₂ concentration.

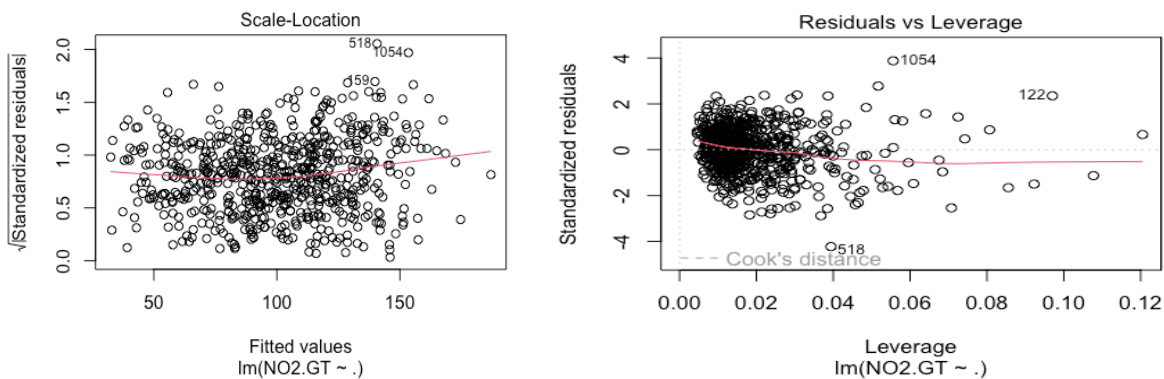
Plots

```
plot(modelLm)
```



Residuals vs Fitted Values plot, each data point represents an observation from the dataset. The x-axis shows the fitted values, which are the predicted values of the response variable (NO2.GT) based on the regression model. The y-axis represents the residuals, which are the differences between the observed values of NO2.GT and the corresponding predicted values.

Q-Q Plot: In a Normal Q-Q Plot, the standardized residuals are plotted on the y-axis, while the theoretical quantiles (NO2.GT) of the standard normal distribution are plotted on the x-axis. That each data point lies on a line indicates that the standardized residuals are approximately normally distributed.



Scale-Location Plot (or Spread-Location Plot): The x-axis represents the fitted values ($\ln(\text{NO2.GT})$), while the y-axis represents the square root of the standardized residuals. scatter data points on the plot and horizontal line on the graph represents that the variance of the residuals is constant across different levels of the predictor variable ($\ln(\text{NO2.GT})$).

Residuals vs Leverage plot: The x-axis represents the leverage, which is a measure of how much an observation influences the model's fit. Clustered data points towards y-axis indicates that certain observations have high leverage but relatively low standardized residuals.

Final model (MODEL-2)

```
lmModel <- lm(NO2.GT ~ CO.GT + PT08.S1.CO + C6H6.GT + PT08.S2.NMHC + NOx.GT +
PT08.S4.NO2 + PT08.S5.O3 + T + RH, data=trainData)
summary(lmModel)
```

```
##
## Call:
## lm(formula = NO2.GT ~ CO.GT + PT08.S1.CO + C6H6.GT + PT08.S2.NMHC +
##      NOx.GT + PT08.S4.NO2 + PT08.S5.O3 + T + RH, data = trainData)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-47.164	-6.882	0.433	7.783	42.018

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.574623	8.699350	0.756	0.4501
CO.GT	11.275055	1.717706	6.564	1.07e-10 ***
PT08.S1.CO	0.050430	0.007419	6.798	2.40e-11 ***
C6H6.GT	-3.143970	0.508800	-6.179	1.13e-09 ***
PT08.S2.NMHC	0.183568	0.011278	16.276	< 2e-16 ***
NOx.GT	0.056470	0.019710	2.865	0.0043 **
PT08.S4.NO2	-0.124044	0.011640	-10.657	< 2e-16 ***
PT08.S5.O3	0.014503	0.003462	4.190	3.18e-05 ***
T	1.284510	0.230825	5.565	3.83e-08 ***
RH	0.370449	0.088863	4.169	3.48e-05 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.27 on 653 degrees of freedom
## Multiple R-squared:  0.8744, Adjusted R-squared:  0.8727
## F-statistic: 505.3 on 9 and 653 DF,  p-value: < 2.2e-16
```

Interpretation of Model Parameters and Fitness

The linear regression model, based on the “**AirQuality**” dataset, aims to predict the concentration of NO2 (Nitrogen Dioxide) includes all available predictor variables in the dataset except the response variable.

Residuals: These are the differences between the observed and predicted values of the response variable (NO2.GT). They represent the model's errors, with a minimum of -47.164, a maximum of 42.018, and quartiles at -6.882 (Q1), 0.433 (median), and 7.783 (Q3).

Coefficients: It estimates the regression coefficients for each predictor variable. They indicate the expected change in NO₂ concentration for a one-unit increase in each predictor variable, holding other variables constant.

Standard Error: This measures the variability of the coefficient estimates. Smaller standard errors indicate more precise estimates.

t value: This is the ratio of the estimated coefficient to its standard error. It assesses the significance of each coefficient. Larger absolute t-values indicate more significant effects.

Multiple R-squared: This is the proportion of variance in NO₂ concentration explained by the model's predictor variables. In this case, approximately 87.44% of the variance in NO₂ concentration is explained by the predictor variables.

In summary, this provides insights into how well the predictor variables explain the variation in NO₂ concentration and the significance of each predictor variable in the model.

Conclusion :

In conclusion, this dataset "AIR QUALITY" provides insights in air quality patterns, especially in urban environments where high pollution levels can have significant implications on public health and environmental sustainability. By analyzing the responses from the gas multisensory device deployed in the Italian city, researchers can gain insights into the dynamics of various air pollutants, including NO₂ concentration.

In this report two models were fitted to analyze NO₂ concentration. The first model specifies these variables, providing a more focused and potentially interpretable model. On the other hand, the second model adopts a more generalized approach by including all available predictors.

Adjusted R-squared of 0.8724 , indicating that approximately 87.24% of the variance in NO₂ concentration can be explained by the predictor variables. This suggests the effectiveness of the models in capturing the complex relationships between sensor responses and pollutant concentrations.

These findings contribute valuable insights into air quality monitoring and prediction, supporting informed decision-making for environmental management and public health initiatives.